

Truck Drilling Machine Requirements		C.Senthilkumar
		Date: 11/01/25
Methods Department	Reference: D / Meth / 050 / Truck Drilling /CH/01 Rev. 1	Page: 1 / 6

1. OBJECTIVE

The objective of this project is to conceive, design, build, assemble and commission a Truck Drilling machine for Clutch Facing.

This facility will allow the drilling of parts in accordance with the Valeo quality standards

2. PRODUCT SPECIFICATION

2.1 Type of product

These are clutch facings made of composite material based on fibers (organic and inorganic), thermosetting resins.

2.2 Dimensional characteristics of product

	Mini	Maxi
Clutch facing Outside Diameter	280 mm	430 mm
Clutch facing Thickness	2.8 mm	6.2 mm
No of Holes in the clutch facing	24 nos	54 nos
Hole diameter	7.75mm	14.5mm
Rivet hole thickness	0.9mm	2.1mm
Rivet hole thickness tolerance	+/-0.1mm	+/-0.1mm
Hole position (Runout) max	0.8mm	0.8mm

2.2.1 Concept diagram.:

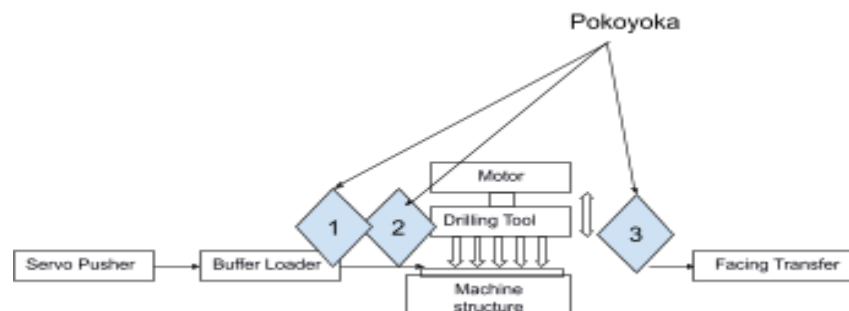


Figure : 1

Truck Drilling Machine Requirements		C.Senthilkumar
		Date: 11/01/25
Methods Department	Reference: D / Meth / 050 / Truck Drilling /CH/01 Rev. 1	Page: 2 / 6

List of other pokayoke based on our past experience from LLC :

- Poka yoke to Check minimum qty in buffer loader
- Poka yoke to Check presence of groove on the top
- Poka yoke to Check centering unit movement or accuracy of locating the facing.
- Poka yoke to check the jig height during drilling to detect overlapping of facings.
- Poka yoke to Check concentricity or offset of the facing 0.8mm max.
- Poka yoke to Check facing exit out of drilling tool.
- Poka yoke to Check facing no hole presence.

2.3 Requirements

2.3.1 Loading area

- The principle of the loading station on the drilling machine to be kept facing.
- This system will allow stocking a minimum of 100-150 parts.
- To detect a minimum level on the stack buffer, a sensor will be installed and a light signal on the operating station.
- If the buffer goes down under the minimum level the drilling machine will stop.
- The loading system will be able to be used for all sizes, facings (diameters, thickness).
- Changing the size can be done using simple settings.
- Pokoyoka to detect the groove position before feeding to drilling
- The operator can load the component by manually without stopping of the machine



Loading area to be provide with suction arrangement to collect dust generated in grinding

Truck Drilling Machine Requirements		C.Senthilkumar
		Date: 11/01/25
Methods Department	Reference: D / Meth / 050 / Truck Drilling /CH/01 Rev. 1	Page: 3 / 6



Maximum stack level 100 to 150 no's

The operator loads the facing in loading area in maximum 100 to 150 nos. The presence of facing in the loading area above the minimum the poka yoke detectors allow to run the cycle when an amount below the minimum in the stack is reached stops the cycle.

3.2.2 Facing Feeding arrangement



➤ It is a "V feeder" which is responsible for pushing the facing in the center of the drill jig.



Component feeding thickness adjustment unit

➤ To control the single facing transfer from Buffer loading to drilling

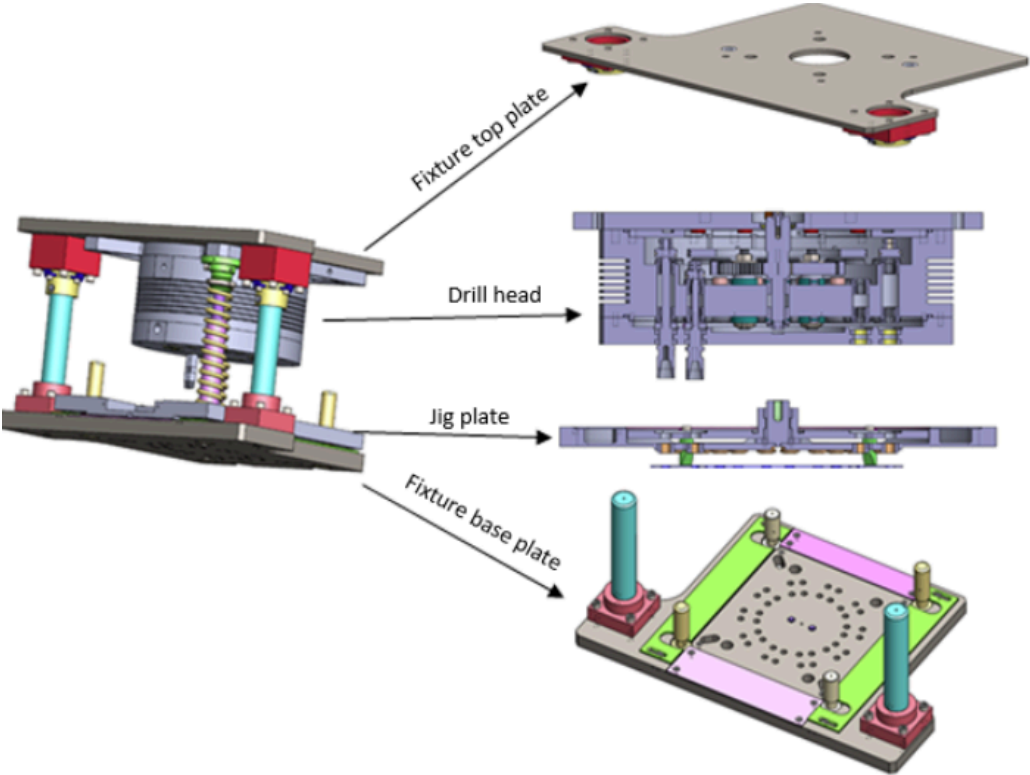
Truck Drilling Machine Requirements		C.Senthilkumar
		Date: 11/01/25
Methods Department	Reference: D / Meth / 050 / Truck Drilling /CH/01 Rev. 1	Page: 4 / 6



Machine bed drilling dust collecting arrangement

- Component drilling dust collecting existing arrangement in the machine bed. For new machine suitable dust collecting arrangement to be considered
- Total tool height 500mm (Existing tool)
- Tool weight 400 kg (Existing Tool)
- Drilling tool clamping and declamping in auto provision
- Tool up down movement and drilling stroke should be in servo control

Existing Drilling Tool Arrangement



Truck Drilling Machine Requirements		C.Senthilkumar
		Date: 11/01/25
Methods Department	Reference: D / Meth / 050 / Truck Drilling /CH/01 Rev. 1	Page: 5 / 6

4. Functions And Acceptance Criteria

Functions	Spécifications	Level	Tolérance
Drilling	Head rotation speed (motor)	1460 RPM	mini
Production flow	Machine capacity	600 p/h	mini
Respect of health and safety norms	Noise level of machine	<75dB	maxi
Ensure a level of reliability	MTBF	8 h	mini
	TRP	85%	mini
Allow for maintenance work fast	MTTR	1 h	maxi
Minimize the drilling layout	Length	Max	3 meters
	Width	Max	2 meters
Ensure accessibility for cleaning and maintenance	Accessibility	Max	-
Detect machine failures	Automatic machine stop	Stop at the first failure	-
Report machine downtime	Light signal	Visibility	-
Identify the stopping causes	Failure display at the operator station	All failures	-
Thickness Under rivet head	Respect to the dimensional tolerance level +/- 0.05	Cp Cpk >1.6	Min
Ensure the maximum eccentricity of the PCD	Respect to the dimensional tolerance	Cmk = 2	mini
Allow adjustment of drilling bits	Accessibility	Maximum	-
Allow to change drilling bits	Accessibility	Maximum	-
Change Over time	Maximum time to change the head	30 min	maxi

Truck Drilling Machine Requirements		C.Senthilkumar
		Date: 11/01/25
Methods Department	Reference: D / Meth / 050 / Truck Drilling /CH/01 Rev. 1	Page: 6 / 6

5.0 General Requirements

- Control panel should IP65 dust proof
- Space should be optimized
- All the motors should be energy efficient (Bharat bijlee/kirloskar/siemens)
- PLC mitsubishi Fx5U series
- HMI proface 10”
- All the safety doors should be interlocked with safety PLC
- MS fabricated Safety door with transparent polycarbonate sheet
- Machine control voltage 24 V
- Machine Safety validation and certification (Decra/Omron/Pilz)
- All other points should be followed Valeo General Equipment Specifications D / Meth / 025 Rév.: 12

5.1 Scope of work

Your offer Must cover :	Supplier	Valeo
Preliminary Process risk analysis	X	
Electric studies	X	
Automation study	X	
Mechanical design	X	
Bill of material	X	
Assembly	X	
Supply of gauges, and components set up and commissioning		X
Set up	X	
Freight packaging	X	
Transport delivery on site	X	
On site commissioning	X	
Power supply connections		X